Nathan Lapp

519-216-8543
nathanglapp@gmail.com • https://nathanlapp.xyz
1506-50 Cambridge Avenue, Toronto, Ontario, M4K 2L3

SUMMARY

Gameplay and systems programmer focused on engine architecture, graphics programming, and interactive systems. Experience shipping a Steam title, building a custom game engine, and developing low-level graphics debugging tools alongside VR/MR gameplay systems.

SELECTED PROJECTS

Realm - Custom Game Engine (D)
https://lappn99.github.io/nathanlapp.xyz/projects/realm.html

- Built a custom game engine in D to explore low-level engine architecture.
- Designed and implemented rendering architecure among several other systems such as an Entity Component System, Asset Loading, and UI

GLTrace - OpenGL Call Tracing & Debugging Tool
https://lappn99.github.io/nathanlapp.xyz/projects/gltrace.html

- Built a two-component OpenGL tracing system consisting of a low-level function hooking layer (C/C++) and a Rust-based analysis and tooling pipeline
- Implemented dynamic OpenGL function interception via API hooking and shared library injection to capture runtime graphics calls from running applications in a non-invasive manner
- Developed a Rust-based tooling backend to process, structure, and create formatted output of OpenGL calls made in a frame

Spider Canyon – 3D Puzzle Platformer (Steam Released)
https://lappn99.github.io/nathanlapp.xyz/projects/spider-canyon.html

- Shipped a complete 3D puzzle-platformer on Steam, developed end-to-end from prototype through release
- Designed and implemented custom player movement and traversal systems, including physics-driven mechanics tailored for skill-based navigation and level design
- Built core gameplay systems supporting iterative tuning, level design workflows, and refinement of movement-based challenges during development

WORK EXPERIENCE

Game Developer Jun 2024 - Present
Invalid Symbol LTD. • Remote

- Architected a modular movement system supporting 20+ abilities for scalable traversal gameplay
- Designed a reusable hierarchical state machine framework used across multiple systems
- Implemented physics-based movement using vector mathematics
- Built JSON-based localization pipeline for text and audio assets
- Implemented VR hand gesture interaction systems for gameplay mechanics
- Led development of a mixed reality game using passthrough occlusion and relighting on Meta Quest

Game Programmer and Consultant Apr 2021 - Sep 2021
Cream Productions • Brampton (Remote), Ontario

- Developed VR gameplay systems and levels for Meta Quest experiences
- Optimized VR performance using GPU profiling and runtime analysis
- Extended animation pipeline and improved usability and runtime performance in Unity

Research Assistant and Programmer Dec 2020 - Apr 2021
Sheridan Institute of Research and Technology • Brampton (Remote), Ontario

- Contributed to digital human facial animation pipeline for VR and virtual production
- Built Unity-based framework integrating with facial animation workflows
- Developed tools using image and video processing techniques for experimental pipelines

SKILLS

Core Areas

- Engine and gameplay systems
- Graphics programming and rendering tools
- VR/MR interaction systems
- Networking and real-time systems
- API design and systems architecture

Technologies

- C#, C, D, Python, GDScript
- Unity, Unreal, Godot
- OpenGL, RenderDoc, Snapdragon Profiler

EDUCATION

B.Sc. Honours, Computer Science - Trent University (2024)
Software Development and Network Engineering - Sheridan College (2021)